STM32F401
DUAL-AXIS SOLAR TRACKER

Project Setup, Wiring, Programming and Operation Guide

Purpose
A practical handover document for students who need to assemble, load, test and operate the working solar-tracker project.

Controller: STM32F401 family
Sensors: 4 LDR voltage-divider inputs
Actuators: 2 servo motors (pan and tilt)
Wireless interface: HC-05 Bluetooth module

Prepared from the supplied project pin-configuration reference. Firmware-specific filenames and exact IDE project settings should be taken from the project source folder.

1. Project Overview

This project implements a dual-axis solar tracker. Four Light Dependent Resistors (LDRs) are arranged in top-left, top-right, bottom-left and bottom-right positions. The STM32F401 reads their analog levels, compares the left and right sides and the top and bottom sides, then adjusts two servos to orient the panel toward the brighter direction.

The HC-05 Bluetooth module provides a simple manual control interface. According to the supplied pin reference, sending **S** or **s** stops automatic tracking; sending **R** or **r** resumes it.

System flow

LDR voltage dividers → STM32 ADC1 → tracking comparison → TIM3 PWM → pan/tilt servos
Bluetooth phone/terminal ↔ HC-05 ↔ USART2

2. Hardware and Software Required

Item	Purpose
STM32F401 development board	Main controller for ADC, PWM and serial communication.
4 LDRs and 4 fixed resistors	Light sensing using four voltage dividers. The reference suggests 10 kΩ as a typical fixed resistor; confirm the actual build.
2 servo motors	Pan (azimuth) and tilt (elevation) movement.
HC-05 Bluetooth module	Wireless serial commands to pause/resume tracking.
Regulated external 5 V supply for servos	Provides servo current without drawing it from the MCU board.
USB/ST-LINK programmer and cable	Programming and debugging the STM32 board.
Keil MDK-ARM / µVision or the IDE used by the supplied project	Open, build and download the existing firmware project.
Bluetooth terminal app	Send S/s and R/r over the HC-05 serial link.

3. Pin Connections

Function	STM32 pin	Peripheral	Connect to
Pan servo signal	PB0	TIM3_CH3 / AF2	Pan servo signal wire
Tilt servo signal	PB1	TIM3_CH4 / AF2	Tilt servo signal wire
Top-left LDR	PA1	ADC1_IN1	TL divider midpoint
Top-right LDR	PA4	ADC1_IN4	TR divider midpoint
Bottom-left LDR	PA5	ADC1_IN5	BL divider midpoint
Bottom-right LDR	PA6	ADC1_IN6	BR divider midpoint
HC-05 TXD → MCU RX	PA3	USART2_RX / AF7	HC-05 TXD
MCU TX → HC-05 RXD	PA2	USART2_TX / AF7	HC-05 RXD

4. Wiring the Hardware

4.1 LDR sensor dividers

Each LDR is connected as a voltage divider. The junction between the LDR and the fixed resistor is connected to its assigned ADC pin. The reference drawing shows 3.3 V → LDR → junction → fixed resistor → GND. Repeat this arrangement for all four sensors, connecting each midpoint to its own ADC input.

Sensor placement: TL = PA1, TR = PA4, BL = PA5 and BR = PA6. Keep the physical sensor positions consistent with the labels used in the firmware.

4.2 Servo motors

Connect the pan servo signal wire to PB0 and the tilt servo signal wire to PB1. Power both servos from a separate regulated 5 V supply. Connect the external supply ground to the STM32 ground so the PWM signal has a common reference.

4.3 HC-05 Bluetooth

HC-05 pin	Connection
VCC	5 V or 3.3 V, depending on the breakout-board specification.
GND	STM32/common ground.
TXD	PA3 (USART2_RX).
RXD	PA2 (USART2_TX).

UART wiring is crossed: module TXD connects to MCU RX, and module RXD connects to MCU TX. The supplied reference states that HC-05 RXD is a 3.3 V logic input and can connect directly to the STM32 PA2 output.

IMPORTANT POWER NOTE: Do not power the servo motors from an STM32 GPIO pin or assume the board's 5 V/3.3 V rail can supply their current. Use a suitable external servo supply and connect grounds together. Check the exact HC-05 breakout-board VCC requirements before powering it.

5. Firmware Configuration Reference

Peripheral / setting	Value in supplied reference
System clock	16 MHz using HSI.
TIM3 prescaler	PSC = 15 (1 MHz timer tick).
TIM3 auto-reload	ARR = 19999, giving a 20 ms period / 50 Hz PWM.
Servo pulse range	1000–2000 µs.
Pan / tilt ranges	Pan: 0–180°; tilt: 0–90°.
ADC1	Software-triggered, single conversion, 12-bit; sample time 84 cycles.
ADC range	0–4095.
Night threshold / deadband	200 / 200.
USART2	9600 baud, 8N1; BRR = 1667; RX interrupt enabled; NVIC priority 2.

6. How to Build and Run the Project

The steps below describe the normal workflow for the existing STM32 firmware project. Use the actual project folder and source filenames supplied by your team; those names are not present in the pin-configuration reference.

Step 1 — Check the hardware before connecting power

• Confirm all four LDR divider midpoints go to the correct ADC pins.
• Confirm pan and tilt signal wires go to PB0 and PB1 respectively.
• Confirm HC-05 TXD → PA3 and HC-05 RXD → PA2.
• Check for short circuits and loose wires.
• Keep the servo supply switched off until the wiring and common ground have been checked.

Step 2 — Open the existing firmware project

• Connect the STM32F401 board to the development computer using the programmer/debugger used by the project (for example, ST-LINK).
• Open the existing project in its configured IDE (Keil µVision if that is the project environment).
• Select the correct target/device and confirm the project uses the STM32F401 variant on the actual board.
• Do not create a new project or change peripheral settings if the supplied working project already builds correctly.

Step 3 — Build the firmware

• Run the IDE's Build/Rebuild command.
• Check the build output for errors. Resolve missing device-pack, startup-file or include-path issues using the project's original configuration rather than guessing settings.
• A successful build should produce the firmware image (commonly an .axf or .hex file, depending on project settings).

Step 4 — Download the firmware to the board

• Ensure the programmer is detected and the target is powered correctly.
• Use the IDE's Download/Flash command to program the STM32F401.
• If the IDE does not start the application automatically, reset the board after programming.
• Keep the servo power disconnected during initial firmware programming and basic board checks.

Step 5 — Power the complete system

• Power the STM32 board using its intended board supply/USB arrangement.
• Power the servos from the separate regulated 5 V supply.
• Ensure the external servo supply ground and STM32 ground are connected together.
• Power the HC-05 according to the module board's voltage specification.

Step 6 — Test automatic tracking

• Place the four LDRs in their TL, TR, BL and BR positions.
• Illuminate one side of the sensor arrangement with a lamp or safely expose it to sunlight.
• Observe whether the pan and tilt servos move to reduce the light difference between the sensor pairs.
• Move the light source gradually and check that the tracker responds smoothly rather than continuously jittering.

Step 7 — Test Bluetooth control

• Pair the phone/computer with the HC-05 using the pairing procedure for the module.
• Open a Bluetooth serial-terminal application and connect to the HC-05 serial service.
• Configure the serial terminal for 9600 baud, 8 data bits, no parity and 1 stop bit (8N1), matching the firmware reference.
• Send **S** or **s** to stop automatic tracking. Send **R** or **r** to resume automatic tracking.

The supplied pin reference documents the command characters and serial settings, but does not specify a particular mobile app or pairing PIN. Use the HC-05 module documentation or the settings used during the

successful project demonstration.

7. Understanding the Tracking Behaviour

The firmware reference defines the two comparison directions as follows:

Azimuth error = average of left sensors – **average of right sensors**
This controls the pan servo on PB0.

Elevation error = average of top sensors – **average of bottom sensors**
This controls the tilt servo on PB1.

A deadband of 200 is specified to avoid unnecessary movement for small differences. A night threshold of 200 is also specified. The exact response when these thresholds are reached depends on the firmware implementation.

8. Testing Checklist

Test	Expected check	Pass
Power and wiring	No hot components, shorts or loose connections; grounds are common.	■
LDR top-left	Changing light at TL changes the PA1 sensor reading.	■
LDR top-right	Changing light at TR changes the PA4 sensor reading.	■
LDR bottom-left	Changing light at BL changes the PA5 sensor reading.	■
LDR bottom-right	Changing light at BR changes the PA6 sensor reading.	■
Pan servo	Responds to left/right light imbalance.	■
Tilt servo	Responds to top/bottom light imbalance.	■
Bluetooth stop	S or s stops automatic tracking.	■
Bluetooth resume	R or r resumes automatic tracking.	■

9. Troubleshooting

Symptom	Checks to perform
Project does not build	Confirm the correct STM32 device pack, startup file, compiler settings and include paths are installed/configured in the original project.
Board is not detected	Check USB cable, ST-LINK connection/driver, target power and debugger selection.
Servo does not move	Check external 5 V servo supply, common ground, signal pin mapping (PB0/PB1), and whether the firmware is generating TIM3 PWM.
Servo jitters or moves unexpectedly	Check sensor orientation, stable power, common ground, wiring and the firmware's deadband/servo limits. Do not force the servo mechanically.
LDR readings do not change	Check the divider wiring, 3.3 V and GND connections, midpoint connection and correct ADC pin assignment.
Bluetooth cannot connect	Check HC-05 power, pairing, serial profile, crossed TX/RX wiring and common ground.
Bluetooth connects but commands do not work	Use 9600 baud, 8N1 and send S/s or R/r. Confirm USART2 RX is connected to PA3 and that the firmware's receive interrupt is enabled.

10. Important Notes for Future Students

• Keep a copy of the original working firmware before making changes.
• Do not change pin assignments without updating the firmware and this wiring guide together.
• Servo motors require a separate suitable power supply; always share ground with the controller.
• Avoid exposing the LDR/ADC inputs to voltages above the STM32 input limits. The reference uses a 3.3 V divider supply.
• Test one subsystem at a time: power, sensors, servo PWM, then Bluetooth.
• Record any changes made to the working hardware, firmware, compiler version or IDE configuration so the next student can reproduce the setup.

11. Reference and Scope

This guide is based on the supplied document *STM32F401 Solar Tracker – Pin Diagram* (4 pages). In particular, the pin map and block diagram are on pages 1–2; sensor, servo and HC-05 wiring plus PWM/ADC/USART settings are on pages 2–4. The step-by-step IDE workflow is general guidance for running an existing STM32 project; the supplied pin reference does not identify the exact firmware filename, IDE project filename, compiler version, flashing configuration, HC-05 pairing PIN or terminal application.

Before distributing this guide, add the actual repository URL, firmware folder/file name, IDE/compiler version and any project-specific setup instructions used in your successful demonstration.